UltraXAI: Technical Strategy

Explainable AI for Tumor and Cancer Detection in Ultrasound Imaging

1. Recommended "Best" Models

For a Final Year Project (FYP) focused on Ultrasound, the model must handle noise (speckle) and low contrast. The "Trending" choice is a **Hybrid Transformer-CNN** approach.

TOP RECOMMENDATION: SEGMENTATION & DETECTION

TransUNet or Swin-Unet

Ultrasound images are textures-heavy. While CNNs are great at local patterns, Transformers capture global context (the relationship between the tumor and surrounding tissue).

- **Why:** It combines the high-resolution spatial details of UNet with the global modeling power of Vision Transformers (ViT).
- **Dataset Fit:** Excellent for BUSI (Breast Ultrasound Images) or Thyroid datasets.

STATE-OF-THE-ART: CLASSIFICATION

EfficientNet-B0/B3 with Attention Gates

If the goal is purely Benign vs. Malignant classification, EfficientNet provides the best accuracy-to-parameter ratio, making it easy to train on standard GPUs.

- **Mechanism:** Adding "Squeeze-and-Excitation" blocks helps the model focus on the actual tumor pixels rather than the black background/ artifacts.

2. The "X" in UltraXAI (Explainability)

Since your title emphasizes **Explainable AI**, the model choice is only half the work. You must integrate XAI frameworks:

Key XAI Techniques to Implement:

- **Grad-CAM / Grad-CAM++:** Visualizes heatmaps showing which parts of the ultrasound image triggered the "Cancer" prediction.
- **SHAP (SHapley Additive exPlanations):** Quantifies the contribution of specific features (like margin irregularity or echo-texture) to the result.
- **LIME:** Explains individual predictions by approximating the model locally with an interpretable one.

3. Comparison Table

Model Architecture	Strength	Difficulty	XAI Compatibility
Attention U-Net	Focuses on tumor boundaries	Medium	High (Attention Maps)
Vision Transformer (ViT)	Captures long-range dependencies	High	High (Self-attention)

Model Architecture	Strength	Difficulty	XAI Compatibility
ResNet-50 + GradCAM	Industry standard, stable	Low	Native Support

4. Suggested FYP Roadmap

1. **Data Prep:** Use the *Breast Ultrasound Images (BUSI)* dataset from Kaggle. Apply speckle noise reduction (Lee/Frost filters).
2. **Model:** Implement **TransUNet**. It is currently very popular in medical imaging research.
3. **Explainability:** Generate heatmaps for every prediction. If the model says "Malignant," the heatmap should highlight the "spiculated margins" of the tumor.
4. **Evaluation:** Use Dice Coefficient (for segmentation) and AUC-ROC (for classification).